

EXAM – CHALLENGE 3: Evaluating Influenza Vaccine Effectiveness Against Mortality: A Causal Inference Approach

Each year seasonal influenza epidemics cause millions of influenza virus infections worldwide, resulting in millions of dollars spent on vaccination programs and treatments of patients. While most individuals who become infected with influenza have mild symptoms and recover without sequelae, some individuals can develop severe illness, particularly those with high risk conditions (e.g., the elderly, pregnant women, and young children). An estimated 3 to 5 million cases of severe illness and 300,000 to 650,000 deaths are caused annually by seasonal influenza epidemics worldwide according to the World Health Organization (WHO). You are tasked with writing a research paper that applies a causal inference framework to examine the effectiveness of the influenza vaccine against influenza-related mortality. Influenza Vaccination has been observed to have different rates of effectiveness, depending on the end-point and the specific populations analyzed. We provide data collected from an observational study in Australia, related to patients who have been admitted to hospitals because of influenza. You are asked to answer the following research questions:

1. What is the effect of Influenza Vaccination on mortality for influenza?
2. Is there any difference in the effect between years when the dominant A subtype of influenza was H1N1 (2010, 2011, 2013, 2014, 2016) and years when the dominant A subtype of influenza was H3N2 (2012, 2015, 2017)?

Part 1: Introduction

Write a brief overview of your research proposal, addressing the following:

- Research Question: Define a clear, focused research question that your study will address.
- Hypothesis: Based on prior findings, state your hypothesis. For example, how do you expect the Influenza Vaccine to affect mortality? Which confounders do you expect?
- Rationale: Explain why this research question is important. Refer to existing controversies in the literature regarding Influenza Vaccine Effectiveness and to relevance of confounders in this setting.

Part 2: Materials and Methods

Develop a basic outline for the research methods you would use, considering the following components:

- Study Population and Sample: Describe the characteristics of your study participants (e.g., number of participants, inclusion criteria, and key demographics), stratified per treatment status and for every stratification you aim to analyze (e.g. the two groups of years)
- Covariates: List relevant covariates you would control for in your analysis with their distributions
- Outcome Measurement: Describe the main outcome you would examine.
- Statistical Models: Specify the type of models and the type of causal inference approach(es) you used, providing rationales for your choices

Part 3: Results

In the Results section of your report, summarize the key findings based on the dataset provided.

Describe any associations you found between Influenza Vaccine and Influenza Mortality. Include coefficient values and confidence intervals, where available, and all relevant metrics.

Part 4: Discussion and Conclusions

Interpret the findings within the causal framework. Address whether the results support the hypothesis that influenza vaccination reduces mortality. Discuss the role of confounding and how it was addressed in the analysis. Highlight limitations, such as potential unmeasured confounding, and their implications for causal inference. Suggest public health implications, such as strategies for improving vaccine coverage or targeting at-risk populations.

Grading Criteria:

1. Clarity and focus of research question and hypothesis
2. Relevance and depth of methodology
3. Appropriateness and rigor of statistical analysis approach
4. Insightfulness of discussion and policy recommendations
5. Coherence between python code and the methodology described in the paper

Dataset Variable Descriptions:

A subset of 1482 patients was selected among those admitted to 17 Australian hospitals between 2010 and 2017 for influenza. Influenza diagnosis was assessed with a laboratory test. Patients were defined as vaccinated if they had received the seasonal influenza vaccine > 14 days before admission. The primary outcome is death from influenza. The following variables have been collected:

1. Age
2. MaleSex (1 if Male, 0 if Female)
3. Year (of admission 2010-2017)
4. Hospital (ID of the hospital of admission 1-17)
5. Vaccinated
6. IndigenousAustralian (1 if belonging to this ethnicity, 0 otherwise)
7. Smoking
8. Obesity
9. Pregnant
10. Homeless
11. NursingHomeResident
12. Diabetes
13. ChronicCardiacDisease
14. ChronicRespiratoryDisease
15. ChronicNeurologicalDisease
16. ChronicRenalDisease
17. LiverDisease
18. Malignancy
19. Immunosuppression
20. FunctionalStatus (in increasing order of severity: Not restricted, MildRestriction, SelfCaring , Limited, BedBound)
21. NumberComorbidities: 0, 1, >=2 (other than those already reported in the variables)
22. Survived (0 means died of influenza, 1 survived to the influenza event)

All comorbidities and risk factors have been assessed at admission time.

Deadline: 15 January 2025 11:00 CET